



SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



**Investigate edge computing in Computer Vision,
defining challenges and presenting solutions.**

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What is Computer Vision?

- Computer Vision helps computers see and understand images and videos.
- It works like human eyes and brain.
- It is used in many smart devices.
- **Examples:**
 - Face unlock in phones.
 - CCTV cameras.
 - Self-driving cars.

WHAT IS EDGE COMPUTING?

- Edge Computing means processing data on the device itself.
- It does not always need the internet.
- It gives faster results.
- **Examples:**
 - Smart cameras.
 - Smartphones.
 - Robots.

HOW DOES IT WORK?

- The camera captures an image.
- The device processes the image.
- The device gives the result immediately.
- **Example:**
 - A smart camera detects a person's face and unlocks the door quickly.

CHALLENGES

Some problems of Edge Computing are:

- Small devices have less power.
- Battery drains quickly.
- Storage space is limited.
- Large AI programs can be slow.

These issues make computer vision difficult on small devices.



SOLUTIONS

We can solve these problems by:

- Using smaller and faster AI models.
- Using better hardware.
- Reducing unnecessary data.
- Saving battery power

ADVANTAGES OF EDGE COMPUTING

- Faster image processing.
- Better user privacy.
- Reduced internet dependency.
- Lower data transmission cost.
- Real-time decision making.
- Edge computing is useful in many smart systems.

CONCLUSION

- Edge Computing makes Computer Vision faster and smarter.
- It helps devices work without depending on the cloud.
- Many problems can be solved with better technology.
- It will be very useful in the future.

THANK YOU!